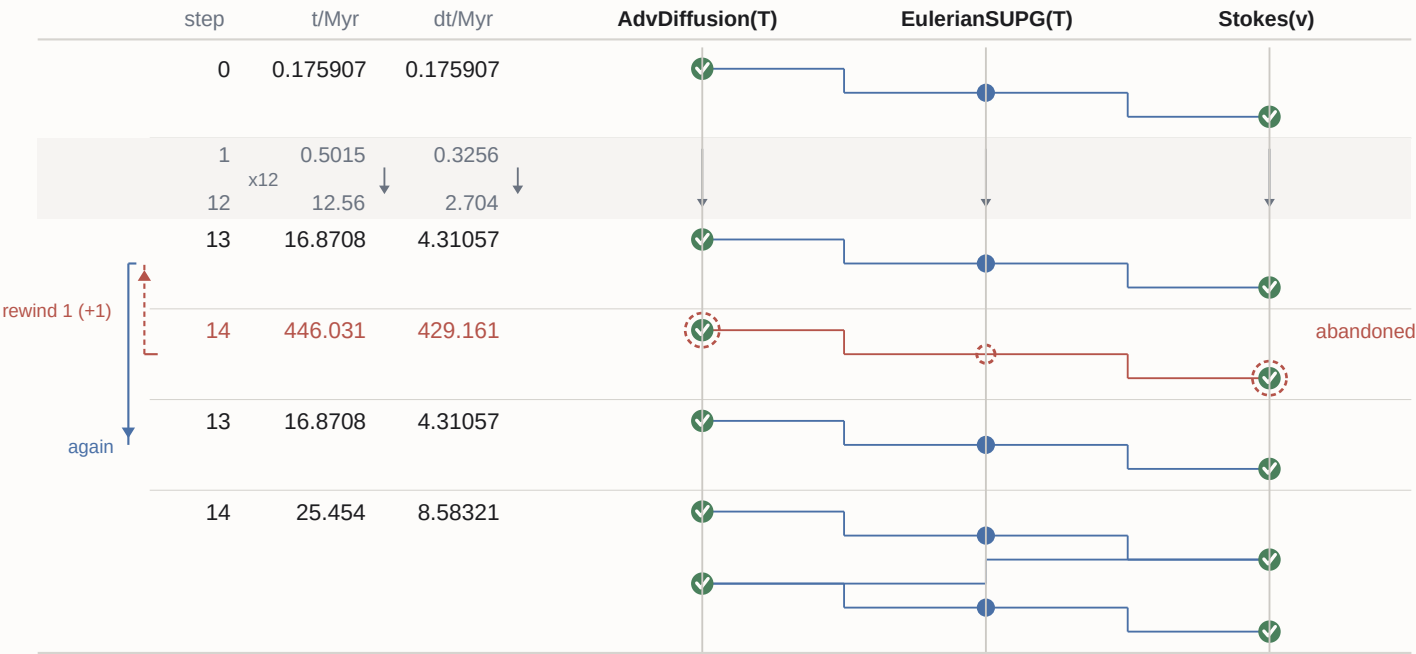


Annulus convection - transcript

started 2026-09-15T13:59:58+10:00 - no terminator: still running, or interrupted - 3 parts



- ran; a step reads down, the path joining what ran in the order it ran
- solved, and converged
- converged, but a fieldsplit block hit its iteration cap - that block did not solve
- did not converge; the reason is in the record
- ran in a step that was then abandoned
- did nothing in this step
- the steps between did exactly this, unchanged; first and last values are shown where they differ

Key - the parts, and what is in their residuals

AdvDiffusion(T)

recorded at step 0

$\Delta t_{18} = 1.147 \times 10^{-6}$ - timestep of this history
 $a_{0,18}^{\text{AM}} = 0.5$ - Adams-Moulton coefficient 0 of history 18
 $a_{1,18}^{\text{AM}} = 0.5$ - Adams-Moulton coefficient 1 of history 18
 $C_{\kappa,18}^{\tau} = 4$ - tau diffusive weight
 $C_{t,18}^{\tau} = 2$ - tau transient weight
 $C_{u,18}^{\tau} = 2$ - tau advective weight
 $\kappa = 10^{-6} \text{ m}^2/\text{s}$ - Diffusivity
 $w_{18}^{\text{SUPG}} = 1$ - SUPG term weight (0 = Galerkin)
essential on Lower: 1
essential on Upper: 0

Stokes(v)

recorded at step 0

$\rho_0 \alpha g = 0.9712 \text{ kg/K/m}^2/\text{s}^2$ - buoyancy coefficient: reference density x thermal expansivity x gravity
 $\eta = 10^{22} \text{ Pa} \cdot \text{s}$ - Shear viscosity
 $\lambda = 0$ - Numerical Penalty
rotated free-slip on Upper: $u \cdot n = 0$
rotated free-slip on Lower: $u \cdot n = 0$